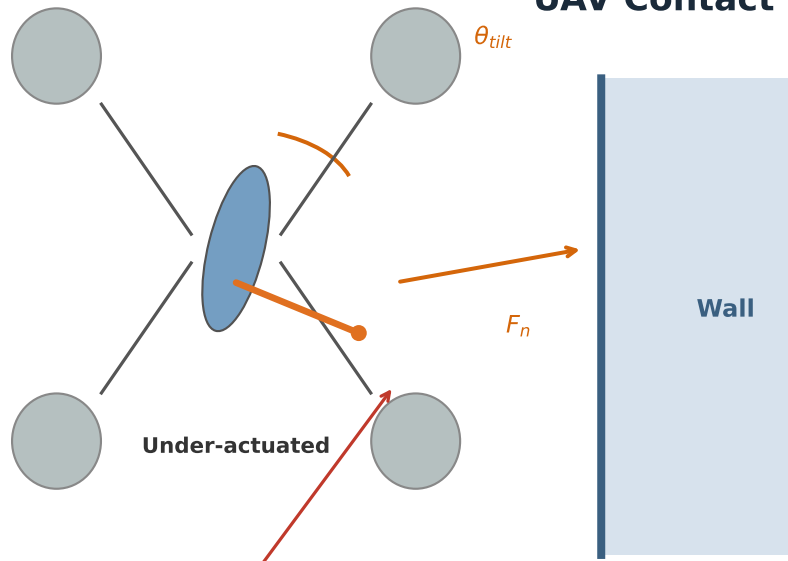
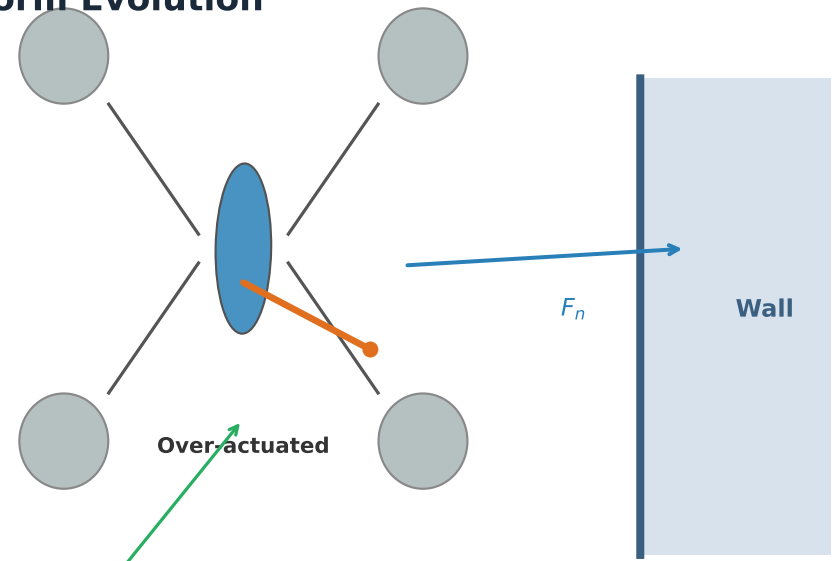


UAV Contact Inspection Platform Evolution



$\theta_{tilt} \Rightarrow \text{probe} \perp \text{surface}$



$\theta_{tilt} \approx 0, \text{probe} \perp \text{surface}$

Passive Compliance
(2019-)

Active Force Control (2019-)

Over-actuated Platform (2020-)

Perception-Control Fusion
(this work)

Under-actuated: tilt-force coupling

Over-actuated: position-attitude decoupled

Mechanical coupling path

Decoupled coupling path